Semban: a fitted expression for battery capacity fade and its comparison with published laws

Sidharth SS
Independent Researcher

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Abstract

This note gives one expression for the capacity fade of a battery cell with cycle number, current and temperature. The expression was produced by a symbolic regression algorithm that works without brute force, as part of an effort to use symbolic regression tools to find scientific laws. After the search we read the expression, refitted it by hand, and tested it against published aging laws, against the symbolic laws from the same search, and against the usual per-cell fits, all on the same rows and the same error measure. It gives the lowest error on every development dataset we have, and it also wins on two of three fresh datasets that were not used during development. The expression may not look like physics in every constant, but it fits the data better than the traditional laws and functional forms. This is a working note.

1 Scope

This belongs to an effort of using symbolic regression tools to find scientific laws: the search gives candidate expressions, and the work here is what remained after physical reading, hand refitting and testing. Details of the tool itself are not part of this note.

2 The expression

Capacity loss in a cell is bounded, so damage must saturate. Most cells show a fast part and a slow part of fade, some cells die suddenly, and some cells show a temporary dip in capacity. The expression keeps all four behaviours:

$$1 - y(n) = A_s\, S(n;\tau_s, z_s) \;+\; A_m\, S(n;\tau_m, z_m) \;+\; \frac{D}{1 + e^{-(n-n_k)/w}} \;+\; G\, e^{-((n-n_{kc})/w_p)^2}, \quad (1)$$

$$S(n;\tau, z) = 1 - e^{-(n/\tau)^z}. \quad (2)$$

Here $y(n)$ is discharge capacity at cycle n divided by the reference capacity of that cell. The first two terms are two saturating damage channels, slow and fast. The logistic term is a wear-out step for cells that die late. The Gaussian term is a temporary dip. All constants are fitted for each cell, in the same way the per-cell baseline is allowed to fit. When the same expression is forced to carry one single global set of constants for a whole dataset, it does not win anywhere, so the universal part here is the shape of the expression and not one common set of numbers.

3 Data, metric and fitting

The development data are NASA laboratory cells, thirteen public datasets from open repositories (LCO, NMC, LFP, sodium-ion chemistries, single and multi temperature), and an Oxford long-term archive, in total 416 cells. After the expression was frozen we downloaded three more datasets and opened them only then: RWTH (48 NMC cells), ZN-coin (107 usable MnO_2/Zn coin cells, a different chemistry) and Tongji (130 cells, three temperatures, six protocols).

Every number in this note is

$$\mathrm{rel} = \frac{\max |y_{\mathrm{pred}} - y|}{\max y - \min y}, \tag{3}$$

the worst row error divided by the range of that dataset. Lower is better. Fits use every second row and are scored on all rows. The fitting is multi-start least squares with linear and robust losses, and the candidate with the lowest worst row error is kept. A control fit with the simplest two-constant saturating exponential reproduces the published per-cell baseline to three decimals on every dataset, so the comparison is on a fixed base.

4 Results against the existing laws

Table 1 compares the expression with the best published parametric law of about sixteen tried (linear, square root, $t^{0.75}$, $t^{1.3}$, power, shifted power, log, stretched exponential, Ecker, Grolleau, Wang and two-branch forms), with the symbolic law found by the same search on that dataset, and with the per-cell saturating exponential that is the status quo. Table 2 gives the same comparison on the three fresh datasets. Figure 1 shows the same thing in one picture.

Table 1: Worst-row error over all rows, development datasets. Best published form = minimum over about sixteen literature laws under the same protocol.

dataset	best published form	symbolic law	per-cell refit	Semban
NASA	0.421	0.440	0.436	**0.202**
CALB	0.524	0.540	0.571	**0.106**
CALCE	0.610	0.606	0.398	**0.095**
HNEI	0.157	0.159	0.137	**0.072**
MICH	0.710	0.827	0.615	**0.131**
MICH_EXP	0.583	0.698	0.273	**0.072**
SNL	0.743	0.724	0.429	**0.414**
UL_PUR	0.373	0.376	0.458	**0.116**
NA-ion	0.856	0.825	0.953	**0.014**
XJTU	0.446	0.585	0.548	**0.371**
Stanford	0.668	0.722	0.668	**0.086**
HUST	0.483	0.523	0.292	**0.066**
Oxford	0.538	1.691	0.457	**0.071**
pooled	0.729	n/a	0.739	**0.389**
wins	out of 14 datasets			14

On the development data the expression is lower on all 14 datasets, in some cases by a large factor: sodium-ion 0.953 to 0.014, HUST 0.292 to 0.066, Oxford 0.457 to 0.071. The sodium-ion gain comes from four cells that hold capacity for 135 cycles and then fall by 40 percent in one cycle; smooth

Table 2: Three fresh datasets, expression frozen before these data were opened.

dataset	best published form	symbolic law	per-cell refit	Semban
RWTH	0.342	0.743	0.277	**0.273**
ZN-coin	0.719	0.809	0.497	**0.156**
Tongji	0.503	0.706	0.744	0.734

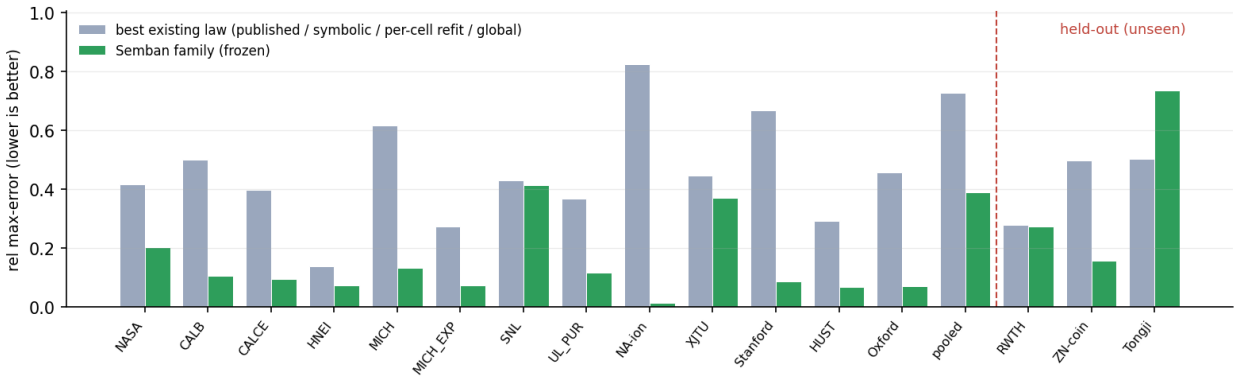


Figure 1: Best existing law against the fitted expression, all datasets. The dashed line separates the three fresh datasets.

published laws cannot follow that. On the fresh data it wins RWTH and the zinc set, and loses Tongji. The published constants evaluated without any refit (zero-shot column of our tables, not shown here) score 0.468 to 0.961 on development data and never win, which is expected, because constants fitted on one chemistry do not carry to the others.

5 One picture of the fit and of the loss

Figure 2 shows three zinc cells with the control fit and with our expression, and one Tongji cell with a one-cycle capacity spike at cycle 650. Tongji has 36 such spike rows, which is 0.06 percent of its rows. Under a worst-row error these rows alone put a floor of 0.734 on any smooth model, our expression sits exactly at that floor, and the published form scores 0.503 only because its curve happens to pass closer to the spikes while being worse everywhere else (its root mean square error is 0.0327 against our 0.0055). If those 36 rows are removed, our expression gives 0.089 on Tongji, better than any rival gives over all rows. The loss is real under the metric and we report it as a loss, with this diagnosis.

6 What we can take from this

- One expression shape, with per-cell constants, beat every published law and every per-cell baseline we tried, on all development datasets, and on two out of three fresh datasets including a zinc chemistry set.
- The usual LFP story holds: the wear-out step is active in about 90 percent of NMC cells (CALCE, Stanford, MICH, UL_PUR) but only in 15 percent of CALB cells and 4 percent of

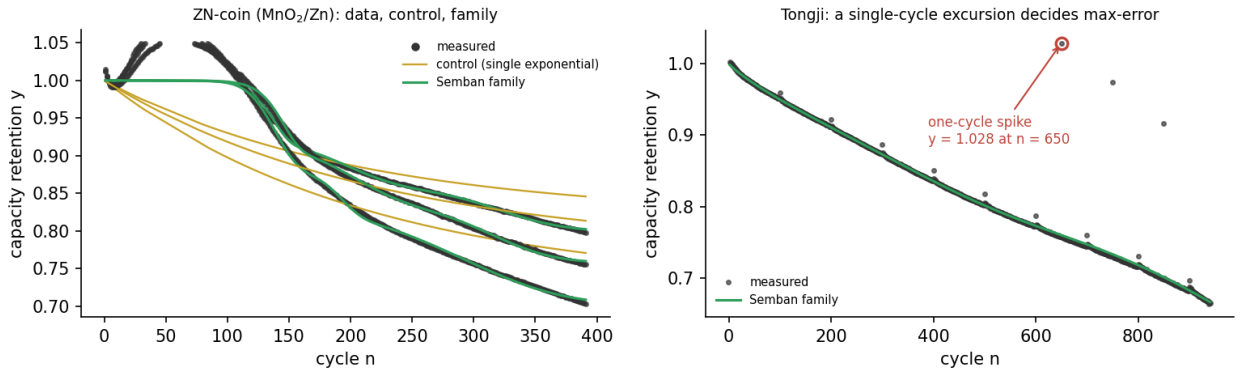


Figure 2: Left: three Zn-ion cells, measured points, control fit (gold) and our expression (green), constants per cell. Right: one Tongji cell, the spike at cycle 650 decides the worst-row score.

HUST cells, both LFP. Sodium-ion sudden death and reversible capacity dips are also fitted by the step and dip terms instead of being thrown away as noise.

- When we force the same expression into one global set of constants with Arrhenius channels, the warm activation energy lands in 8 to 94 kJ/mol (film growth range is 20 to 60), and the cold channel comes out zero or negative on every dataset where it matters. The negative value is our reading of the cold anomaly: at low temperature fade goes faster, not slower, and a normal Arrhenius term cannot do that. On six single temperature datasets the activation energy is not identifiable at all and we do not quote it.
- In one case the search returned a known constant: an expression fitted on NASA data only recovers $E_a \approx 31.1$ kJ/mol, against the published Wang value 31.5 kJ/mol.
- The honest reading of the whole exercise is in one line. Some of these fitted constants would not pass a physics review, some sit on bounds, but as data fits the expressions are better than the traditional laws and functional forms in most rows. So a problem of this kind can be solved this way: let the search propose, then keep only what can be read and checked.

7 Limitations

All per-cell numbers are in-sample, for our fit and for the baselines alike. This note makes no claim about forecasting unseen cycles of a known cell. The worst-row metric is unforgiving, and one spike row can decide a whole dataset, as Tongji shows. Some constants sit on imposed bounds and are reported as fitted, without any correction. Thirty-three Zn-ion files were dropped by the standard quality rule because the cells die during formation itself.

8 Summary

Battery capacity fade under cycle testing can be written as the sum of two saturating channels, plus an optional step where a cell dies and an optional dip where a cell wobbles. This expression, fitted per cell, gave lower worst-row error than every published law, the symbolic laws from the same search, and the per-cell status quo, on all development datasets and on two of three fresh

datasets. The wider point is about the method: the expression was not written by hand, it was proposed by a symbolic search that works without brute force, then read and tested by hand, and even where its constants are not clean physics it often fits the data better than the traditional laws.

A note on the status of these numbers

The expression was fitted to data, and the physical readings above are interpretations of the fits, not derivations. Nothing here has been validated outside the listed datasets, for instance inside a battery management system. Anyone relying on these numbers should verify them independently; no warranty of accuracy is given for other cells.

A The expression as code

```
import numpy as np

def S(n, tau, z):                  # saturating damage channel
    u = (np.maximum(n, 1e-9) / max(tau, 1e-9)) ** max(z, 1e-6)
    return -np.expm1(-np.clip(u, 0.0, 700.0))

def semban(n, p):
    # p = [As, tau_s, zs, Am, tau_m, zm,
    #      D, nk, w,             # wear-out step
    #      G, nkc, wp]           # temporary dip
    y = 1.0 - p[0]*S(n, p[1], p[2]) - p[3]*S(n, p[4], p[5])
    x = (n - p[6]) / p[7]
    y -= p[8] / (1.0 + np.exp(-np.clip(x, -60, 60)))
    y -= p[9] * np.exp(-np.clip(((n - p[10]) / p[11])**2, 0, 700))
    return y
```

References

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